Leftover Bytes

(Problem to Product - Final Pitch)

Team Members (Product Managers)

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From Problem to Solution

Problem at Boston University

- Leftover food from campus events frequently goes to waste.
- The issue is **not due to lack of demand,** but **due to poor communication.**
- Informal Slack posts by staff are **inconsistent** and **often fail to reach students.**
- **Budget-conscious students** and those who want to **reduce food waste** often miss these opportunities.

Our Question?

"What if there were a faster, more structured way to connect leftover food with the students who need or want it?"

Introducing Leftover Bytes

- **Mobile-first platform** designed for quick and easy access.
- Sends **real-time notifications** about available food nearby.
- Enables **one-tap claiming of leftover food portions.**
- Allows **students to update food availability and its status.**
- **Gamifies participation** through points, leaderboards, and rewards to drive engagement.

Product Vision & Strategy

Vision

- **Mobile-first, real-time platform** to redistribute leftover food from campus events.
- **Minimize food waste** through structured, student-driven coordination.
- Focus on equitable, **first-come, first-serve access** – not limited to food-insecure students.

Strategy

- Deploy the MVP at **CDS-hosted events** to gather behavioral insights and optimize the user experience.
- Let students **filter alerts based on dietary needs** (e.g., vegan, gluten-free) to ensure relevance.
- Students receive **instant notification** with food description, dietary tags, photo, location, and timestamp – all tailored to their preferences.
- By **uploading real-time photos** of available food portions, students can update the food status and earn points to boost their engagement on the platform.
- Drive participation through points, **certificates, and leaderboards.**

User Research Summary

Participant Sources

- 12 BU Students (Interviews)
- Stakeholders (Zero Waste Manager, CAS Events)
- 94 Students responses (Google Forms)
- 2 Event Organizers (BU Spark)

What we heard : User Voices

"Sometimes I'd go and the food was already gone. If people updated the status, that'd help."

"I wish I could preview the food and see when it was posted before deciding to walk there."

"Posting needs to take under a minute. Otherwise I won't do it."

"I'm vegan, so filters are important. Otherwise I waste time going for food I can't eat."

"Rewards make it fun to contribute - I'd totally post photos if there were goodies involved."

"I'd prefer a quick posting process, don't want to wait around for students to pick up the food after posting."

"First-come, first-serve makes it simple – once the food is posted, it's fair game for anyone nearby. We don't have the time to manage reservations or hold items."

Insights from User Feedback

Students

- **Prefers a mobile app** over websites or Slack channels.
- Strong preference for **real-time, dietary-filtered alerts.**
- **Direct in-app photo uploads** only (no external uploads).
- Value rewards like **certificates and goodies,** and **prefer anonymous leaderboard participation** (e.g., "SnackSavior123").
- Want to feel part of BU's green community by **contributing and reducing waste.**

Organizers

- Favor **quick upload process** (photo + dietary tags) — minimal effort.
- **Don't want to manually distribute food.**
- Would use a **simple dashboard** for event analytics.
- **Reduce the amount of food wasted** after events.

Stakeholders

- Endorse **First-Come, First-Serve pickup approach** for simplicity and scalability.
- Recommend **partnerships with BU Spark and Dining Services** to scale the platform.
- Support using waste **dashboards for food waste tracking to measure impact.**

Minimum Viable Product (MVP)

Student Login (Kerberos)

Leverages BU Kerberos credentials to auto-authenticate students and organizers securely.

Dietary Preference Setup

Uses manual tagging of uploaded food items from organizer photos using check boxes to match against user preferences: vegan, gluten-free, halal, nut-free, etc

Smart Push Notifications

Uses dietary preferences to send out notifications when food becomes available, reducing irrelevant alerts.

Event Activation System

Organizers upload pictures with time and location tagging; students can preview food before heading over.

Gamified Feedback & Status Updates

Students can upload new pics, or leave freshness notes to earn eco-points in an anonymous leaderboard in exchange for rewards like certificates or sustainability badges.

Dashboard for Organizers

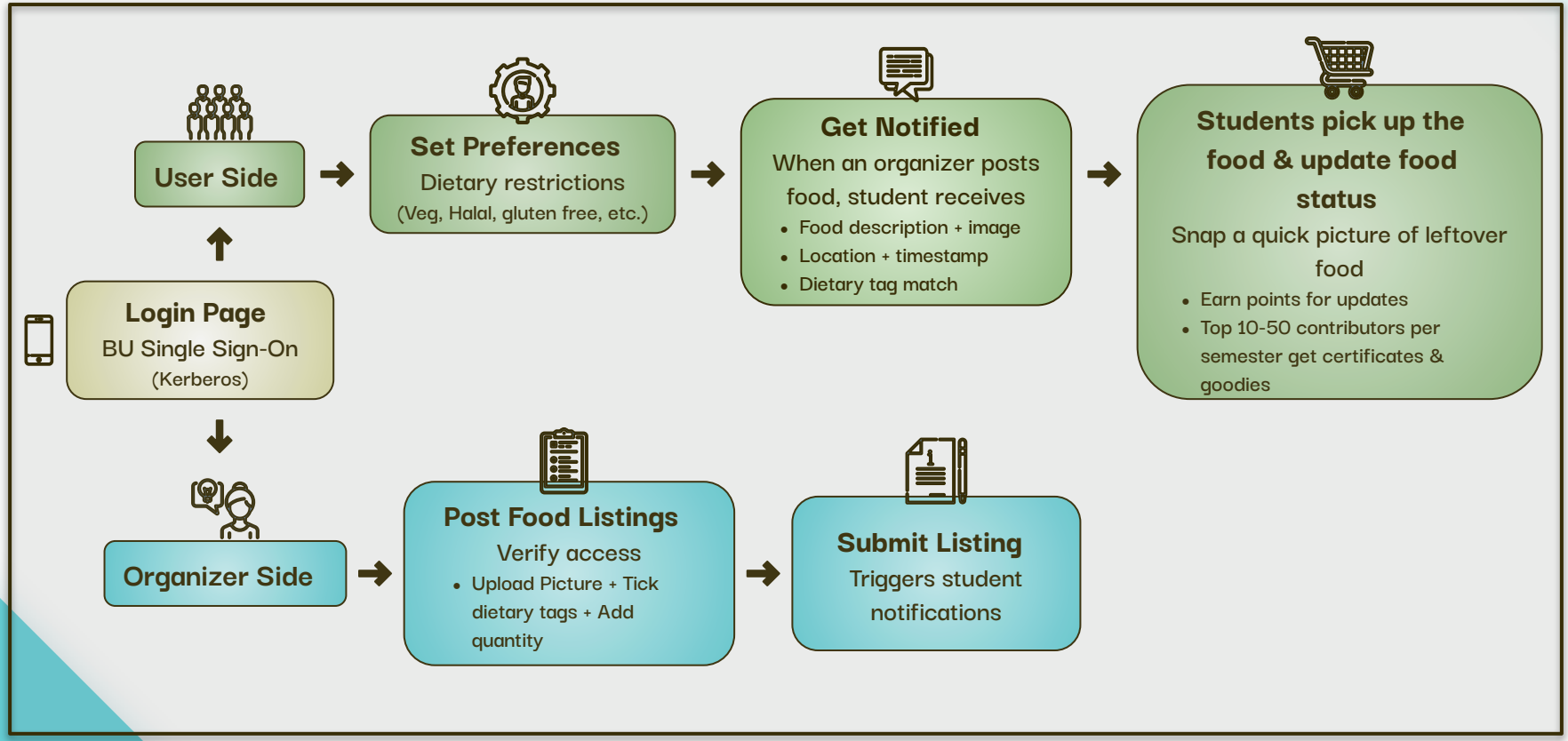
Post-event analytics show how many portions were picked up, waste prevented (in lbs), and impact score (visible to Zero Waste Office).

MVP Features: Launch vs Deferred

✅ Launch-Ready Features
BU Kerberos login
Dietary preference setup & filtering
Organizer food uploads + manual food-type tagging
Real-time push notifications
Preview of food & location is sent before pickup.
Student earn points by marking food as picked.
Leaderboard + Certifications for active contributors
Flag/report stale food posts
Simplified organizer flow (Image → Tag → Location)

❌ Deferred Features
AI-powered auto-tagging
NLP classifies food items based on users dietary preferences
Social media sharing
QR code-based food verification
Pickup time tracking
Real-time counters for status monitoring
Admin tools and multi-campus support

Flowchart


User Side
Set Preferences
Dietary restrictions
(Veg, Halal, gluten free, etc.)
Get Notified
When an organizer posts food, student receives
Food description + image
Location + timestamp
Dietary tag match
Students pick up the food & update food status
Snap a quick picture of leftover food
Earn points for updates
Top 10-50 contributors per semester get certificates & goodies
Login Page
BU Single Sign-On
(Kerberos)
Organizer Side
Post Food Listings
Verify access
Upload Picture + Tick dietary tags + Add quantity
Submit Listing
Triggers student notifications

RICE Framework

Features	R (Reach)	I (Impact)	C (Confidence)	E (Effort)	Score
1 Dietary Filters	8	9	9	3	216.0
2 Student Uploads & Status	8	8	9	3	192.0
3 Push Notification	7	8	9	3	168.0
4 Points & Leaderboard	6	7	8	4	126.0
5 Organizer Dashboard	5	6	8	4	96.0

$$\text{RICE Score} = \frac{\text{Reach} \times \text{Impact} \times \text{Confidence}}{\text{Effort}}$$

(Higher RICE Score → Higher Build Priority)

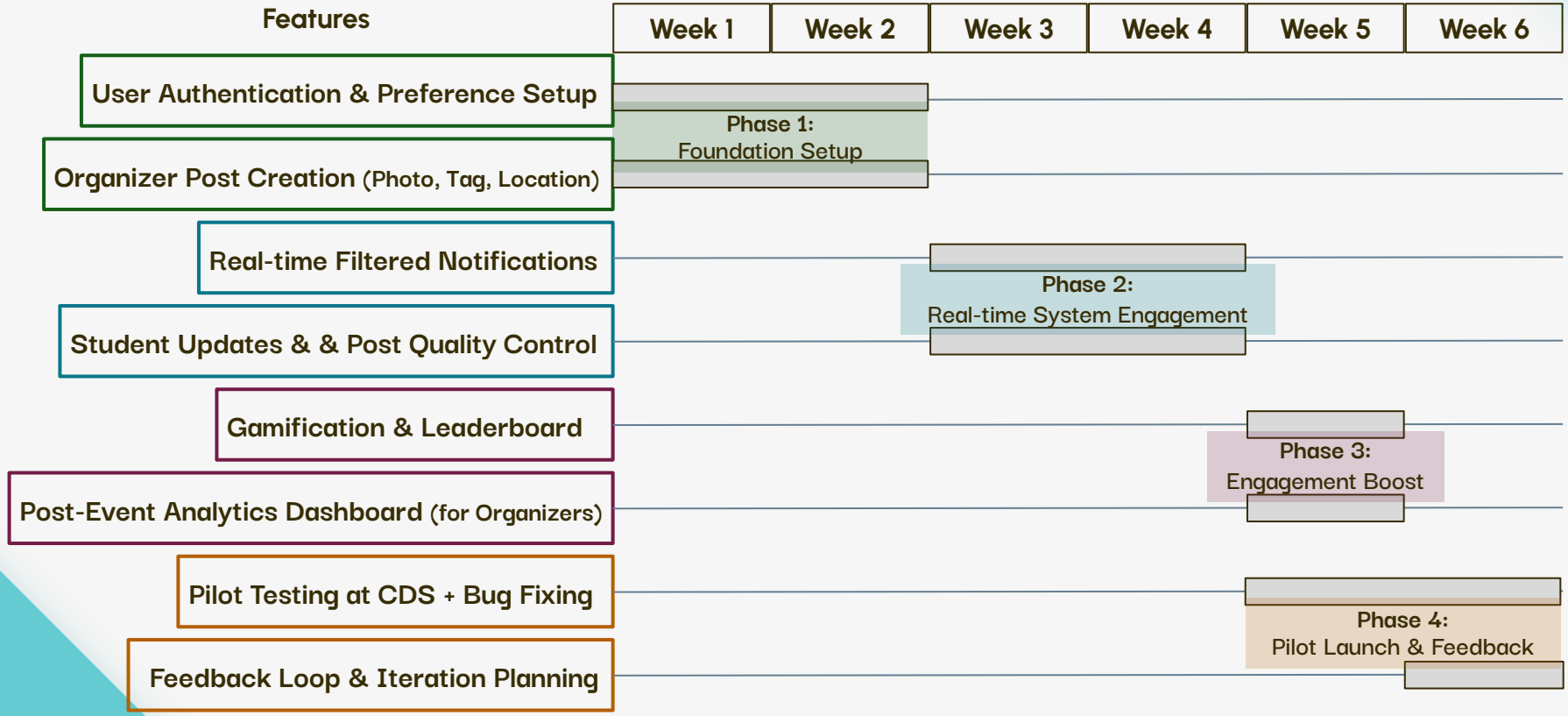
- **Reach** - Number of users impacted per week (students, organizers).
- **Impact** - How much it helps achieve our mission (reduce food waste, boost engagement).
- **Confidence** - How certain we are of success, based on feedback, surveys, and interviews.
- **Effort** - Estimated team-weeks needed to design, build, and test the feature.

Technical Scoping: Technologies & Roles

Layer	Technologies
Frontend	Flutter or React Native (for MVP mobile app)
Backend	Firebase (Authentication + Firestore) (for MVP database and user management)
AI APIs (Future Phase)	Google Cloud Vision / AWS Rekognition (for automated food categorization)
Geolocation (Future Phase)	GPS + WiFi Triangulation (for smart, location-based alerts)

Role	Responsibility
Mobile Developer	Build Android/iOS app (MVP features)
Backend Developer	Setup database, authentication, and notifications (MVP)
UI/UX Designer	Design wireframes, user flows, and visual branding (MVP)
Product Manager (PM)	Manage coordination, sprints, and task tracking
Research Lead	Collect user feedback, run surveys, and track NPS

Agile Development Timeline



Development Roadmap

Phase	Timeline	Feature	Rationale	Impact
Phase 1: Foundation Setup	Weeks 1-2	User Authentication & Preference Setup	Ensure secure access (BU Kerberos) and personalized notifications through dietary filters	Secure student logins and deliver relevant food alerts
	Weeks 1-2	Organizer Post Creation (Photo, Tag, Location)	Allow event organizers to quickly post available food with minimal friction	Drives fast food postings, initiates food availability
Phase 2: Real-time System Activation	Weeks 3-4	Real-time Filtered Notifications	Ensure students receive timely food notifications based on dietary needs	Improves pickup success rate and student engagement
	Weeks 3-4	Student Updates & Post Quality Control	Enable real-time updates on food status (picked up, still available, freshness), allow flagging of stale posts	Reduces food waste and maintains trustworthiness
Phase 3: Engagement Boost	Week 5	Gamification and Leaderboard	Incentivize consistent student participation in anonymous leaderboards via points, certificates, goodies	Boosts engagement, encourages repeat usage
	Week 5	Post-Event Analytics Dashboard (for Organizers)	Help organizers track event impact (pickups, waste reduction)	Strengthens organizer retention and event contributions
Phase 4: Pilot launch & Feedback	Weeks 5-6	Pilot Testing at CDS + Bug Fixing	Test MVP live, identify technical issues, gather real-world feedback	Validates MVP usability and prepares for broader launch
	Week 6	Feedback Loop & Iteration Planning	Collect user feedback during pilot, prioritize improvements, and plan next sprint	Guides product refinement and ensures user-driven evolution

Success Metrics vs North-Star Metric

Success Metrics

The **broad set of goals** that define if your product is doing well.

- **Pickup Success Rate**
  Percentage of posted food portions that are successfully picked up by students.
- **Drive Student Engagement**
  Foster active participation in posting, picking, and updating food status.
- **Deliver Real-Time Alerts**
  Provide students with fast, dietary-relevant and location-based food notifications
- **Maintain Platform Reliability**
  Ensure smooth app performance with minimal bugs and high notification success

North-Star Metrics

The **ONE single metric** that best captures the **core value your product** delivers to users. It aligns the whole team.

- **Food Waste Reduced Percentage (%)**
  Redistribute leftover food efficiently at campus events..

Phase-1 (Pilot Phase)

$$\text{Food Waste Reduced Percentage(\%)} = \frac{\text{Food waste before app} - \text{In week}_i}{\text{Food waste before app}} \times 100$$

Phase-2 (Development Phase)

$$\text{Food Waste Reduced Percentage(\%)} = \frac{\text{Food waste In week}_{i-1} - \text{In week}_i}{\text{Food waste In week}_{i-1}} \times 100$$

- Where i is the week we are currently in after app launch.
- The information about food wasted before the app was given to us by stakeholders

Launch Metrics

- Launch Metrics - Basic technical + operational health check when you first launch the app.

Pickup Success Rate (%)

How much posted food is actually picked up

(Picked Up ÷ Posted Portions) × 100

Average Time to Pick Up

Average time between posting and pickup

(Pickup Time – Post Time)

Active Users

Users posting, picking, or updating weekly

Count of active users per week

Onboarding Completion (%)

% of users who complete signup & dietary preferences

(Completed ÷ Started Signups) × 100

Notification Delivery Success (%)

% of notifications delivered without failure

(Delivered ÷ Sent Notifications) × 100

Bug Reports & Resolution Rate (%)

% of bugs resolved out of bugs found

(Resolved ÷ Found Bugs) × 100

Short-Term KPIs

- Short-term KPI - Business and user behavior metrics to validate if your product is working (engagement, adoption, food saving).

Metrics	Description
★ **Food Waste Reduced** (lbs/week)	Start tracking estimated pounds of food saved weekly at CDS pilot events
Active Engagement Rate	% of students who post, update, or pick food weekly
Pickup Success Proxy Rate	% of food posts with at least one student engagement
Average Time to First Engagement	Time between notification sent and first action taken
Organizer Adoption Rate	% of events where organizers post food

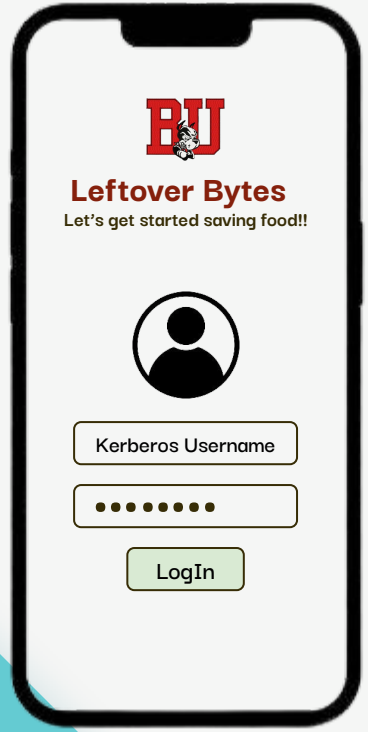
Long-Term KPIs

- Long-term KPI - Strategic goals that show whether the product is succeeding over months (loyalty, scale, repeat use).

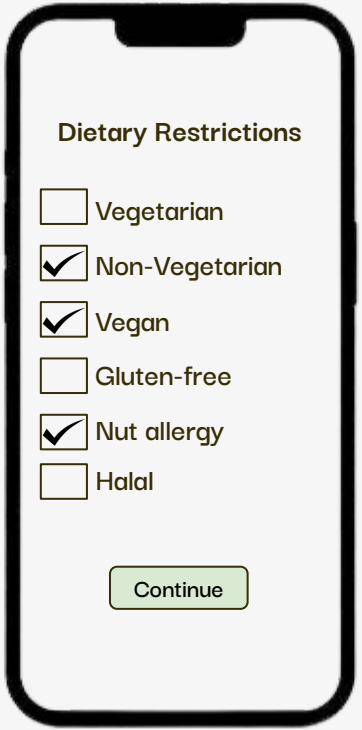
Metrics	Goal
★ Food Waste Reduced (lbs/week)	Maintain/scale to 100–150 lbs/week
30-Day Repeat User Rate	> 40%
Net Promoter Score (NPS)	> 45
Event Coverage Rate	≥ 75%
Community Contribution Rate	≥ 50% event contribution rate

Wireframe: BU Students

Login Page

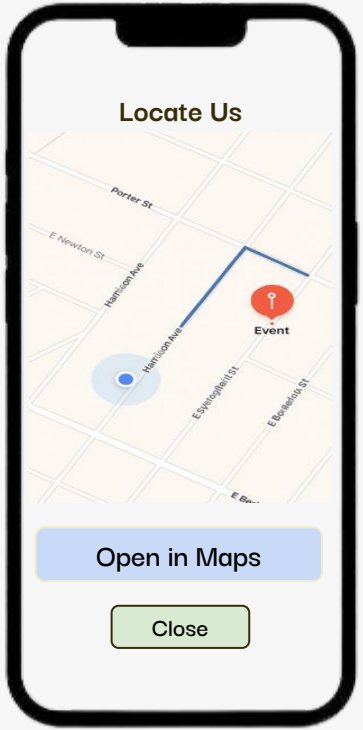


Preferences



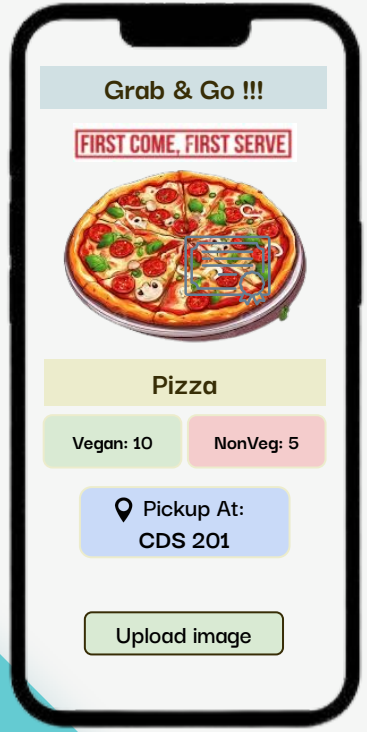
Food Listings

Location



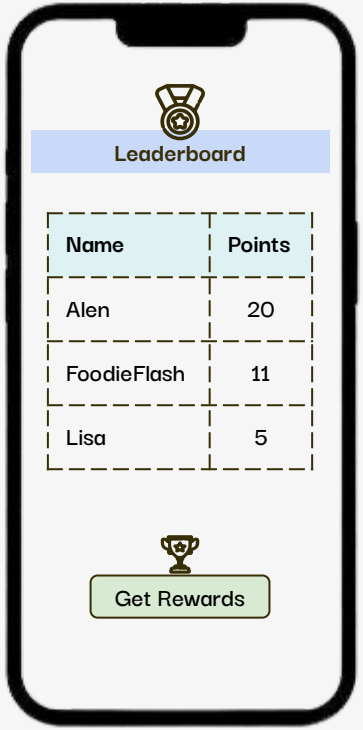
Wireframe: BU Students

Claim Your Meal



Upload Image

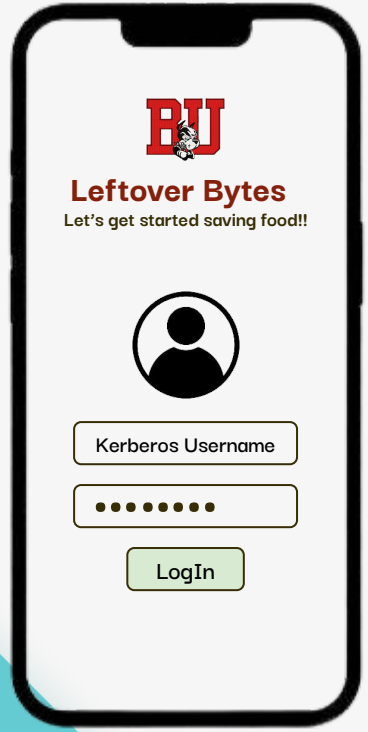
Leaderboard



Rewards Page

Wireframe: Event Organizers

Login Page



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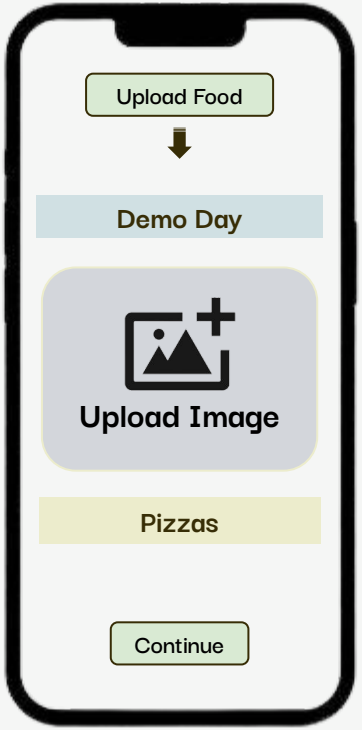

Preferences

Events Conformation

User Feedback on Wireframe

Student Side

- **Kerberos login and dietary preference** setup are **intuitive and match user expectations.**
- Food listings are visually **engaging and emphasize key info** (e.g., titles like **“FREE Pizzas!”**).
- **Gamification features** – points, leaderboards, and rewards – **strongly motivate users.**
- **Direct photo uploads** and clean claim flow support **fast interactions.**

Organizer Side

- **Minimal posting flow** (Photo → Dietary Tags → Location) aligns with organizer need for speed.
- **Post confirmation screen** provides clarity and encourages continued participation.

Risks and Mitigation Strategies

Risk	Reason	Mitigation Strategy
Food Safety & Liability	Potential illness from old/cold food	Add disclaimers, freshness tags, timestamps, safety tips
Student Data Privacy	Handling location & dietary data	BU Kerberos login, opt-in filters, Firebase encryption
Organizer Drop-Off	If organizers stop posting, app fails	Simplify posting flow, show impact stats, provide incentives
Low Student Adoption	Without traction, no pickup activity	Gamify participation, partner with orgs, ambassador rewards
Stale or False Listings	Students waste time going for unavailable food	Add status updates, timestamps, flagging option

Leftover Bytes connects students, cuts food waste, and builds a strong BU Community – and we're ready to launch!

Thank You!